CS 350 – From Deep Learning To Data Mining

EcoSort: CNN-Based Waste Classification Model

Kathy Saldana Renteria
Kylie Ueda

April 21st, 2026

Introduction

Research Problem:

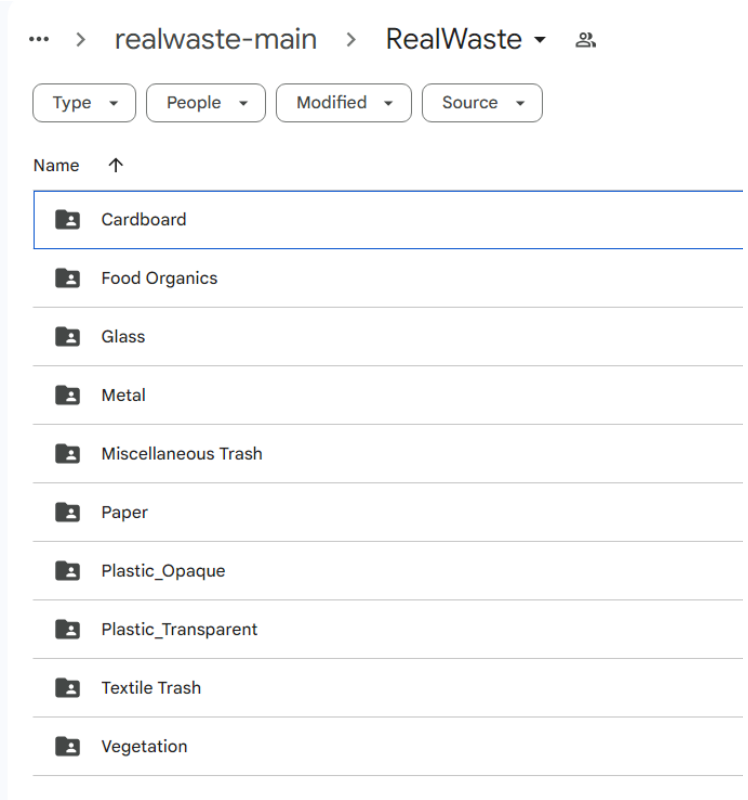
- Waste is growing globally, causing pollution and greenhouse gas emissions
- Manual sorting is slow and error-prone; **CNNs** are often trained on clean datasets, not real landfill images
- Misclassifying waste increases methane emissions

Goal:

- Build an accurate **waste classification model** addressing class imbalance
- Add educational feedback on environmental impact

Dataset: **RealWaste**

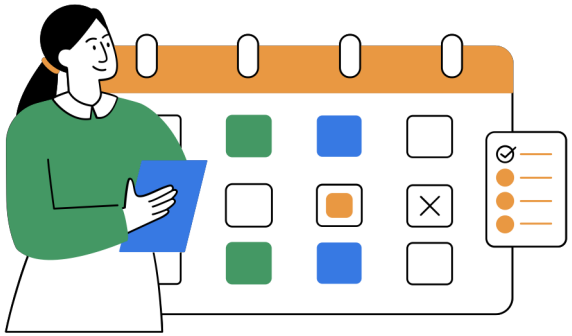
- **4,752 images**
- Image Classification Task
- Captured at an actual landfill reception point
- More realistic than studio-based datasets
- **9** original classes (expanded to **10** with plastic split)

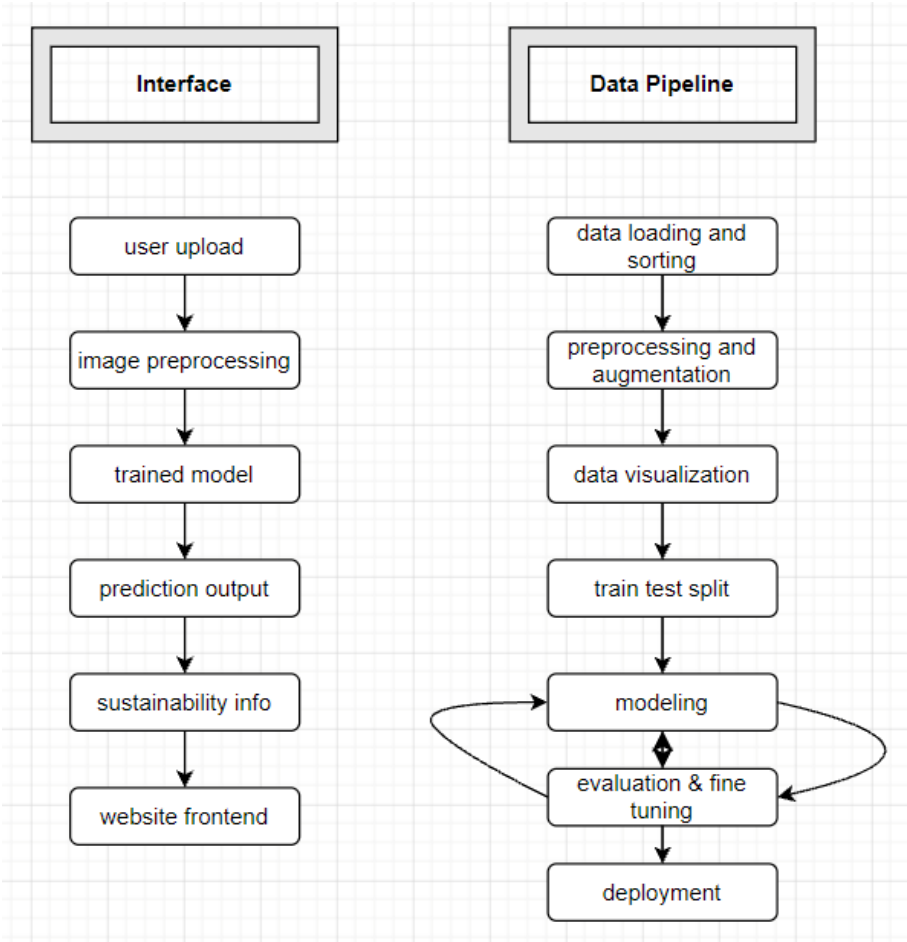


Background

Model/System	Description	Advantages	Challenges
Traditional ML (SVM, KNN, Decision Tree)	Early waste classification approaches using handcrafted features and **classical machine learning algorithms**.	**Simple**, interpretable, low computational cost.	**Poor performance** on complex images, heavily dependent on manual feature engineering, not scalable for real-world variability.
CNN & Transfer Learning (ResNet, VGG, MobileNet)	**Deep learning-based** image classification models often use pretrained **architectures** to improve performance.	**High accuracy**, automatic feature extraction, and effective even with limited data.	**Sensitive to dataset imbalance and noise**, requires large and diverse datasets for best performance.
Real-World & Advanced Systems (EfficientNet, RealWaste dataset)	**Modern approaches** using optimized CNNs and real-world datasets, sometimes integrated with automation.	**Strong real-world performance**, better generalization, applicable to practical waste sorting systems.	**High system complexity**, data inconsistency, increased computational and implementation cost.

Data Pipeline





Technologies Used

Data Loading & Sorting: pandas, os, shutil, glob, google.colab.drive

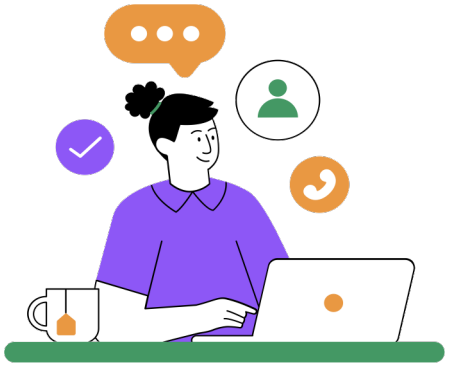
Preprocessing & Augmentation: tensorflow.keras.layers (RandomFlip, etc.), image_dataset_from_directory

Data Visualization: matplotlib.pyplot, seaborn

Modeling: tensorflow.keras.applications

Evaluation & Fine Tuning: classification _report, metrics, ModelCheckPoint, EarlyStopping

Deployment: HTML, CSS, Flask, flask-cors



Preprocessing

Data Cleaning

- Plastic Classes Refined
- Dataset Structuring
  - Manual split
- Image Preprocessing
  - Resized to 224 x 224

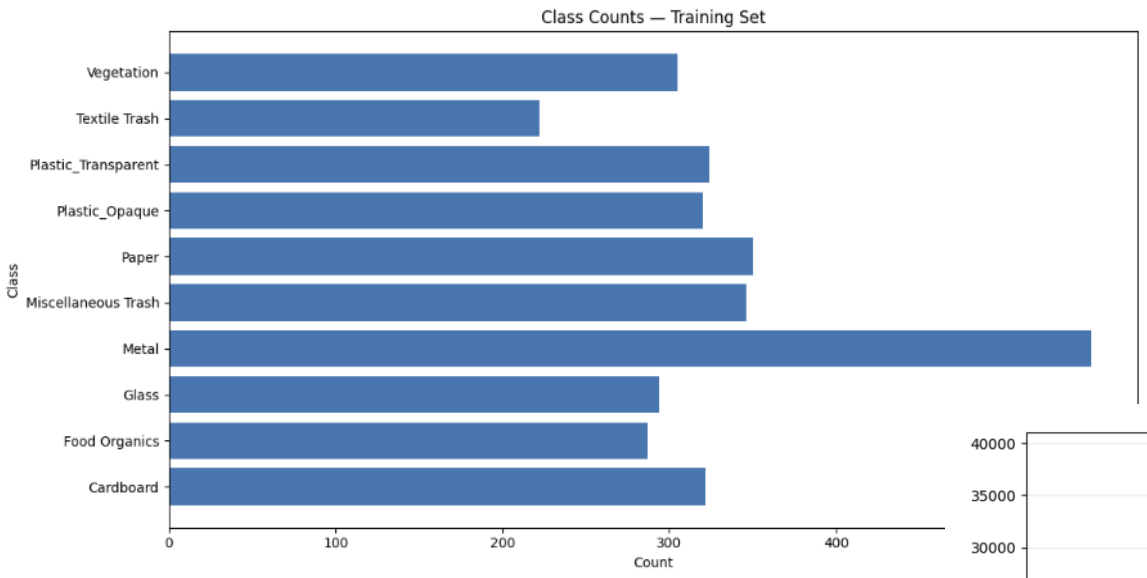
Class Imbalance

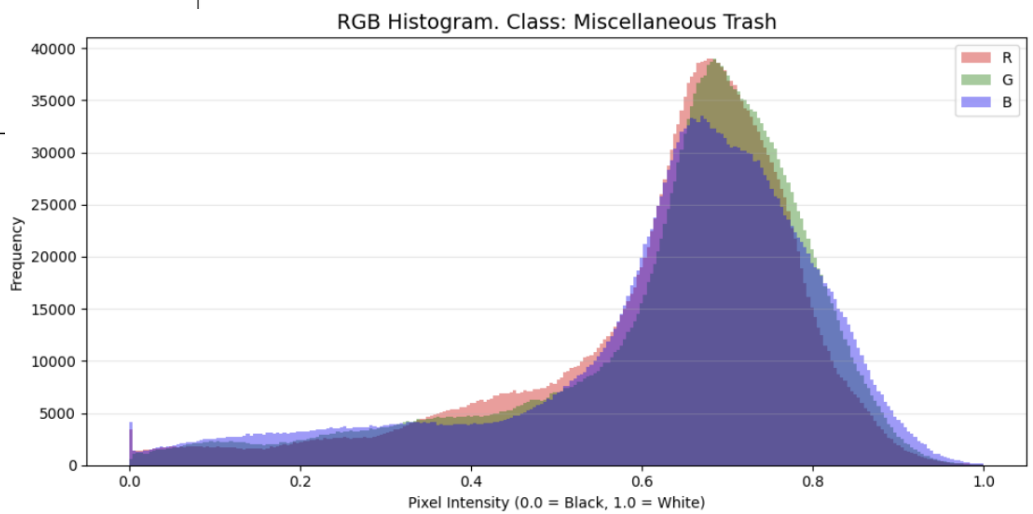
- Model Bias and Low Recall
- Class Weighting Added
  - Penalizes mistakes more heavily

Augmentation

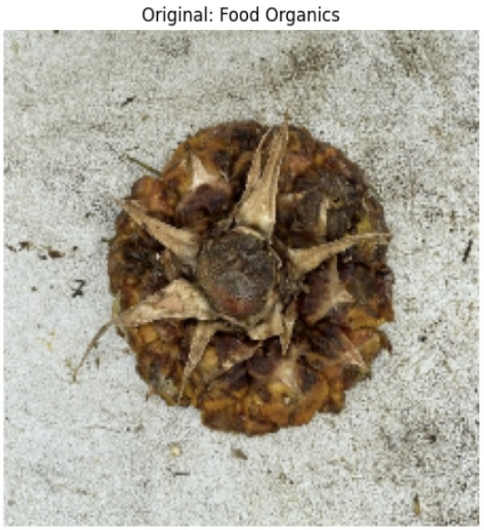
- Applied to Training Set Only
  - Flips (H/V), Rotation (0.15), Zoom (0.1), Brightness & Contrast (0.1)

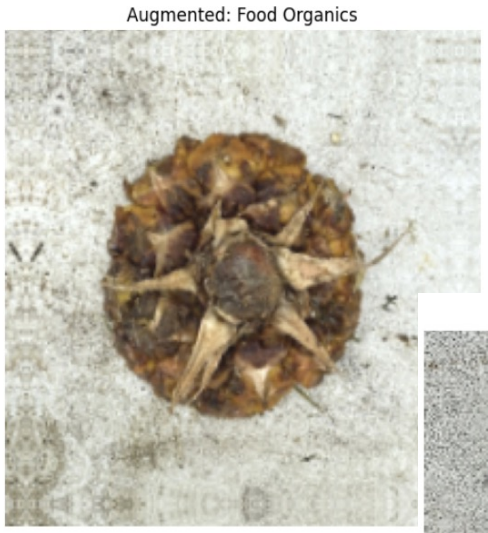
Data Visualization

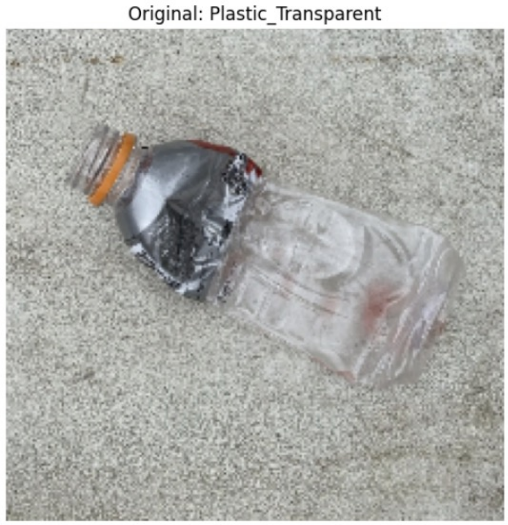




Data Visualization







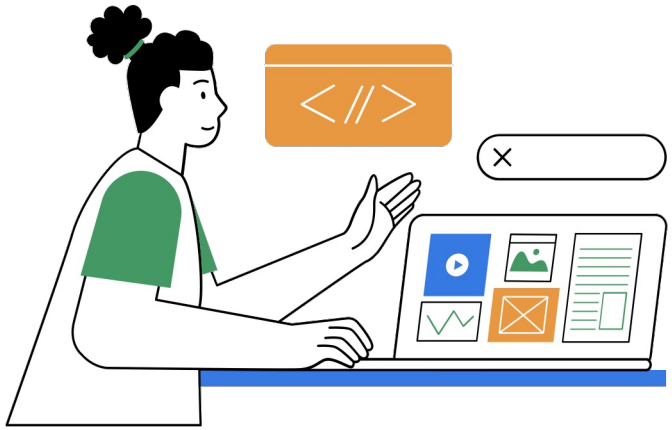
Modeling

MobileNetV2: Lightweight & Deployable **(80.42%)**

- ~4% lower accuracy but ~40% smaller with respect to parameters
- Uses inverted residuals and depth wise convolutions to reduce computation and support performance

DenseNet121: Strong Feature Reuse **(84.03%)**

- Only 0.69% below EfficientNetB0
- Uses dense connections (each layer connects to all previous)
- Promotes feature reuse and efficient gradient flow



Modeling

EfficientNetB0: Best Overall Balance **(84.72%)**

- Higher accuracy with relatively low parameters
- Uses compound scaling (balances depth/layers, width/channels, resolution/image size)

EfficientNetB7: Highest Accuracy **(85.14%)**

- Achieves the top accuracy among all models
- Larger model with significantly more parameters than EfficientNetB0
- Uses compound scaling

Hyperparameter Tuning

Class Weights
(553 vs 222 images)

- Balances Metal vs Textile Trash
- Improves minority class recall

Early Stopping
(patient = 3)

- Stops when validation loss stops improving
- Restores best weights

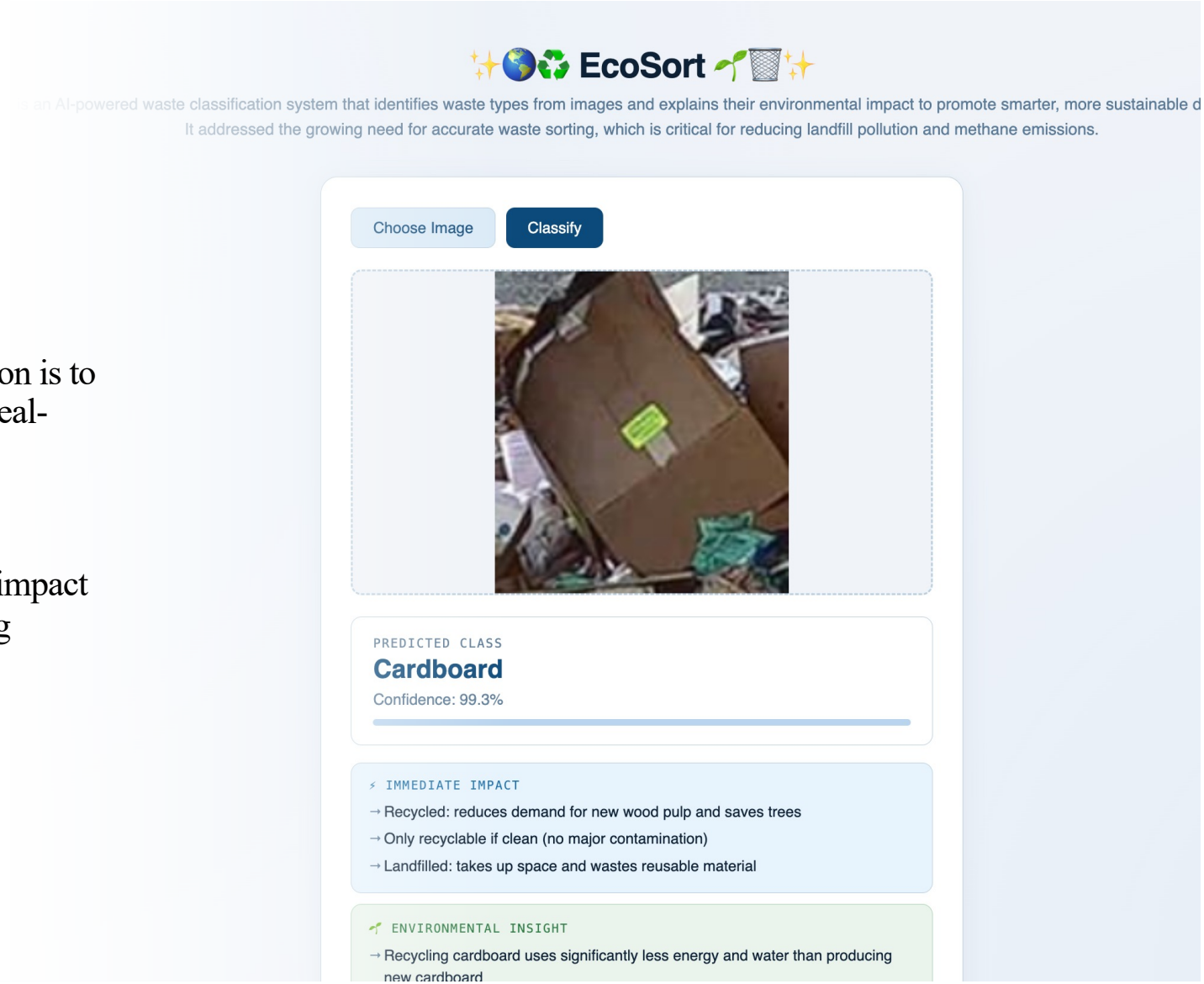
Learning Rate
(1e-4 -> 1e-6)

- Large updates early, small later
- Protects pretrained features

Application

- The purpose of this application is to use a CNN model to classify real-world waste images.

- It aims to educate users by explaining the environmental impact of misclassification, promoting sustainable disposal decisions.



Thank You!

- **CS350: FROM DATA MINING TO DEEP LEARNING**
- saldanarenteriak@etown.edu
- uedak@etown.edu